Natural Ingenuity

Aliaa Abdallah
Amany Yasser
Habiba Sadek
Samar Mohamed

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Abstract

Egypt faces significant challenges due to climate change. By 2050 the temperature will have risen by 1.5°C to 2°C. This warming exacerbates water scarcity, with an estimated 20% reduction in Nile River flow expected both impacting the country's agriculture and food security. This project contributes to decreasing climate change by building a natural smart house operated by renewable energy. In order to have air exchange the house is built in shape of cuboid and its roof is pyramidal . The row materials that give a building strength are palm fronds and petioles. A coating of clay was applied inside the house since the heat conductivity of clay is low. The raw materials of the house and its design are beneficial to lower the interior temperature of the house by 3°C over the exterior temperature. Whole-house control is provided by a smart system. It regulates the humidity in the house by measuring its percentage. The humidity inside the house varies between 35% to 40%. Normal air has a relative humidity of 30% to 40%. Humidity is inversely proportional to temperature. The LDR (Light sensor) controls the light intensity of the LED which makes the summation of sunlight and LED intensity inside the house 225 LUX (Lumens per meter squared). The renewable energy source utilized to power home is solar energy. The goal of a solar tracker is to maximize solar energy harvesting. A servo motor is used in this system to move the solar panels towards areas of higher light intensity, and two LDR sensors are used to monitor variations in light intensity.

Introduction

Egypt faces many grand challenges, including recycling garbage and waste for economic and environmental purposes, increasing the industrial and agricultural bases of Egypt, addressing and reducing pollution of the air, water, and soil, and increasing the technological and scientific bases are the main problems for capstone this year. Constructing a building using eco-friendly material will address and reduce pollution of the air, water, and soil. Constructing a smart building is the best solution for increasing technological bases like Singapore Smart City, yet it has a disadvantage, that it is hard to be applied in Egypt due to its high cost. Also, there is a challenge to follow up on the temperature inside the building to know if the materials have reduced by at least 3°C.(Abderrahim et al.,2022) The selection of palm fronds and palm petioles stems from their intrinsic properties such as water resistance, insulation, and shading attributes. (Lomertwala et al., 2019) In addition, clay was chosen because it is sustainable in all production stages from raw material extraction to production and packaging. It also has low thermal conductivity. The building system includes a light sensor, a temperature sensor, and a humidifier. A temperature sensor was used for precise thermal monitoring both internally and externally. A light sensor was used to adjust the light intensity of the building. A humidifier was used to control the humidity levels. A solar tracker was employed to convert sunlight into electricity and use it for charging batteries. Design requirements emphasize parameters like humidity which has a range of 35% to 40%, temperature which decreased by 3°C from outside, and summation of light intensity inside the house is 225 LUX. The measuring ruler was used to determine the internal dimensions of the building, revealing it to be 54 cm in length, 54 cm in width, and 63 cm in height. Its pyramid-shaped roof, standing at 10 cm in height and matching the base dimensions of the building, not only boasts exceptional wind resistance but also facilitates an efficient ventilation system. This design is particularly favored in storm-prone regions for its structural stability.

Materials and Methods

Materials:

Table (1): List of materials used

Item	Figure	Function	Item	Figure	Function	Item	Figure	Function
Palm Petiole		Constructing the walls of the building	Palm Fronts		Constructing the walls of the building	Aswan clay		Covering the interior of the building
DHT11 Humidity Temperature sensor		Measuring the temperature and humidity	Breadboard 830 point (MP-102)		Connecting the electronic components together	Photo-Resistor Sensor (LDR 5mm)		Measuring the light intensity within the building
Mini Micro Servo Motor SG90 180 Degree		Solar tracking	Resistors ¼ watt		Controlling the flow of current within the LEDs and photo-resistors	Bluetooth Module (HC-05)		wireless communi-cation between electronic devices using Bluetooth
Humidifier		Increasing the humidity within the building	Arduino UNO R3		Controlling the system	Solar Panels		Charging the batteries
2Rechargeable Batteries Li-ion (7.4v)		Energy storage	Jumpers		Connecting the electronic components together	BMS 1S (5v-1A)		Charging and protecting Li-ion batteries
Arduino USB Cable		Connecting the Arduino UNO to the laptop	PIR Motion Sensor Module		Detecting the motion of the door	LEDs 5mm		Lighting the building

Methods:

Safety Precautions:

As shown in Figure (1), safety procedures were followed while building the house in the fab lab under the supervision of the Capstone leader.

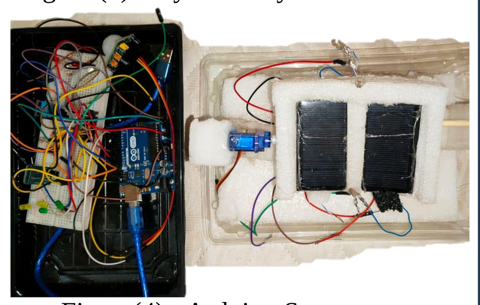
Figure(1) : Safety Precautions

Constructing the prototype:

First, As shown in Figure (2), the house was built using palm fronds, with total inner volume of 0.1982 m^3 measuring 54 cm in width, and length; and 63 cm in height. The palm was collected and punched with a drill, and each wall and roof was built separately. The parts were then assembled, and the house was completed. Palm fronds were twisted around palm petioles. Third, as shown in Figure (3), the house's interior was covered with clay to make it more heat-insulated. Second, As shown in Figure (4), the house's system includes a motion sensor that turns on and off the LEDs (the first actuator used in the prototype), a PIR (motion sensor) that measures and controls the light intensity of light, a DHT-11(Temperature and Humidity sensor) that measures the temperature and humidity of the house, a humidifier that releases cold air, a solar tracker that causes the solar panels to follow the direction of light, an HC-05 (Bluetooth module) that sends data to a mobile application, and an LDR(light sensor) that measures and controls light intensity . The solar tracking was applied using a servo motor (the second actuator used in the prototype). Third, after connecting the sensors, As shown in Figure (5),the code was created in the Arduino UNO application using the C++ programming language. Third, an Android Bluetooth controller was installed to receive data from the Bluetooth module. Fourth, the system's electricity comes from solar energy. The solar panels (10 volts) charge the battery (7.4 volts), which then supplies electricity to the Arduino UNO.

Figure (2): Lateral view of house

Figure (3): Layer of clay inside house



Figure(4): Arduino System

Test Plan:

Three test plans were done. The first test included observing the temperature interior and exterior of the house. The second test involved measuring the humidity within the house by using a DHT-11sensor every three days for 13 days; the results were detected by temperature and humidity sensor. The humidity was 39%. The third test involved measuring and controlling the concentration of the LEDs within the home at different times of the day utilizing PIR (Motion sensor module) and LDR (Light sensor).

Figure(5): QR code or the project codes

Results

Table(2):Relation between humidity & time (days)

Days	Humidity within the building (%)
Day 1	39
Day 2	38
Day 3	39
Day 4	40
Day 5	39

Table(3):Relation between temperature inside and outside & time (days)

Days	Temperature within the building (°C)	Temperature outside of the building (°C)
1st day	34	37
2nd day	25	29
3rd day	23	26
4th day	35	40
5th day	25	28

Table(4):Relation between light intensity &light brightness& time (days)

Hours	Light Intensity (LUX)	Light Brightness (LUX)
10:00 AM	142	83
1:00 PM	177	48
4:00 PM	220	5
7:00 PM	109	116
10:00 PM	69	156

Analysis

The mentioned results were collected by meeting the design requirements by decreasing the temperature within the building by 3°C than the outside, to adjust the humidity within the building to reach the relative humidity in normal air which is in the range between (30%-40%), in addition to stabilizing the light intensity within the building to be 225 LUX (Lumens per meter squared). As shown in Figure(6), after measuring the temperature inside and outside the building every 3 days for 13 days, it was found that the inner temperature of the building was less than the outer temperature by at least 3°C. It was achieved due to the insulation of building materials, the thickness of the walls, and the positions and sizes of ventilation openings. In addition to the high height of the building, which increased the effectiveness of heat convection within the building.

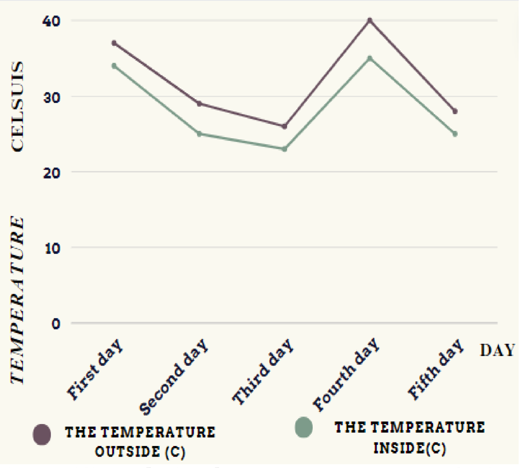


Figure (6): Relation between temperature & time

After the humidity inside the building was measured for 13 days (once every 3 days), the prototype succeeded in collecting the results shown in Figure (7). The results were reached due to using palm fronds, and petioles as construction materials.

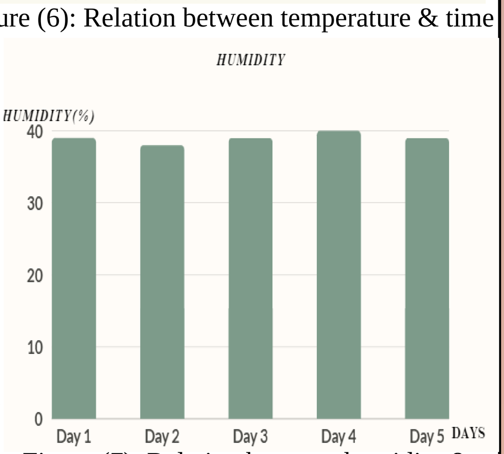


Figure (7): Relation between humidity & time

As shown in Figure(8), the light intensity within the building and the brightness of the LEDs were measured 5 times during the day. The addressed results were reached using the code shown through the QR code in Figure (10), which states that the brightness of the LEDs was reached after subtracting the light intensity within the building from 225 LUX which is the maximum brightness of a LED. The equations RL=500/lux, and Vo=5*RL/(RL+3.3), where RL is the resistance of the LDR sensor, Vo the output voltage of a sensor, and LUX is the light intensity, were used in the LDR sensor to calculate the light intensity. Solar tracking was used to increase the effectiveness of solar energy generation, by increasing the amount of sunlight received during the day. The inner volume of the building was calculated using the following equation: Inner volume = 1.259 h^3*10^-3 m^3, where the width, length of the building equal h cm, and the height of the building equals 1.168 h cm. the dimensions of the building were determined using the golden ratio (1:1:1.168), since it creates an organic, balanced, and aesthetically pleasing composition. The roof of the building is in pyramid shape; because it's stable, especially against the wind because the attack is small. The equations

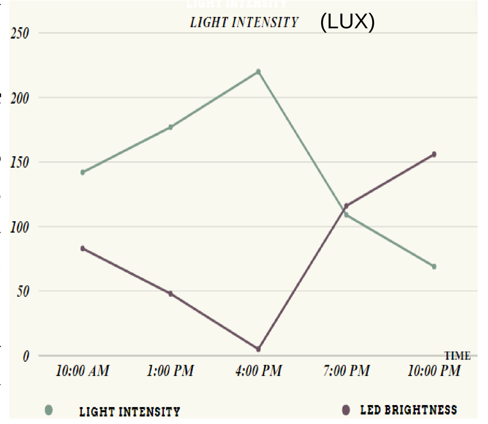


Figure (8): Relation between light intensity and time

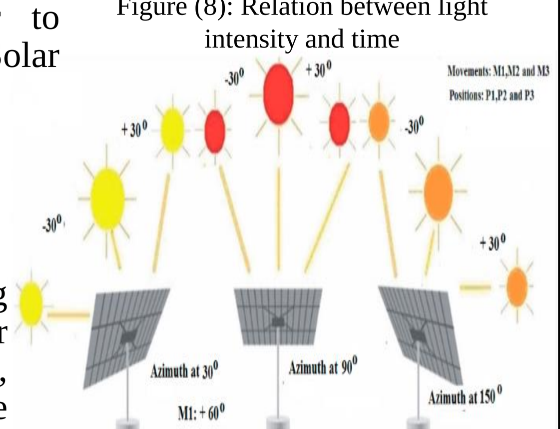


Figure (9): Solar Tracking Mechanism

Figure(10): Motion sensor QR code

Learning Outcomes:

BI.2.09: positive and negative feedback in the hormone cycle in the female and male reproductive system was so helpful in knowing the mechanism of each sensor. **CH.2.11**: knowing the effects of the combustion of organic compounds on the environment, was helpful in choosing completely natural material for the building. **MA.2.06:** finding a relation between the height and the volume of the building using limits. **CH.2.09**: Use a rechargeable battery such as a lithium battery with a certain power that can operate the system and know how to charge. **ME.2.06**: knowing the center of mass of regular shapes helped in maintaining the stability of the building. **CS.2.08:** variables, data, types, assignments, debugger errors, variable scope, loops, and conditions were used during the coding of Arduino, sensors, and actuators in the prototype

Conclusion

To conclude, a palm petiole was used as the skeleton for the house's construction, and palm fronds were twisted around it. Inside, the home was painted with Aswan clay. these row materials of the building achieve the design requirements. The temperature was decreased by 3°C inside the building than outside, and the humidity is between 35% and 40% which offers healthy air inside the building. The total light intensity in the house the whole day is a constant value which is 225 LUX (Lumens per meter squared). This design requirement was achieved by the window in the front face of the house which helps to light the house naturally. When the light intensity detected by the LDR sensor changed, the LED brightness changed to keep the summation of the LED brightness and the intensity of the natural light is 225 LUX (Lumens per meter squared).

Recommendations

Some recommendations were chosen to enhance sustainability and performance. This includes using a DHT22 sensor for higher accuracy and integrated temperature measurement, alongside applying natural additives like straw or fibers to clay for adding strength and reducing cracking. Multiple layers of clay may be employed to create a thicker, more durable insulation barrier. Additionally, a modular green roof system may be also designed to complement palm tree leaf belts, offering extra insulation, and effective rainwater management. In the smart city, each building will contain solar panels that connect to huge batteries that provide the whole city with electricity. This project will be built in Marsa Allam. These strategies collectively aim to improve building resilience, energy efficiency, and environmental impact.

Literature Cited

- Abderrahim, B., Amine, T., Abdellah, M., & Ahmed, H. A. M. (2022). Thermal and economic evaluation of new insulation materials for building envelope based on date palm waste. *2022 2nd International Conference on Innovative Research in Applied Science, Engineering and Technology, IRASET 2022*.
- Cosentino, L., Fernandes, J., & Mateus, R. (2023). A Review of Natural Bio-Based Insulation Materials. In *Energies* (Vol. 16, Issue 12). MDPI.
- Dakheel, J. A., Del Pero, C., Aste, N., & Leonforte, F. (2020). Smart buildings features and key performance indicators: A review. Sustainable Cities and Society, 61, 102328.
- Hoang-Tien, N. (2020). *Climate Change and Sustainable Architecture in Smart Cities*.
- Ismail, kamal, Lino, F., Henriquez, J., Teggar, M., Laouer, A., Arici, M., Benhorma, A., & Rodríguez, D. (2023). Enhancement Techniques for the Reduction of Heating and Cooling Loads in Buildings: A Review. *Journal of Energy and Power Technology*, *05*(04), 1–44.
- Lomertwala, H. M., Njoroge, P. W., Opiyo, S. A., & Ptoton, B. M. (2019). Characterization of Clays from selected sites for Refractory Application. *International Journal of Scientific and Research Publications (IJSRP)*, *9*(11), p9581.
- Mohammed, A. J., Abood, E. A., Jalal, M. A., & Ibrahim, I. K. (2023). Mechanical and Thermal Properties of Polyurethane-Palm Fronds Ash Composites. *Metallurgical and Materials Engineering*, *29*(2), 23–35.
- Neufert, Ernst., Neufert, Peter., & Kister, J. (2012). *Architects' data*. Wiley-Blackwell.

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For Further information

Aliaa.1122572@stemmaadi.moe.edu.eg
Amany.1122509@stemmaadi.moe.edu.eg
Habiba.1122526@stemmaadi.moe.edu.eg
Samar.1122562@stemmaadi.moe.edu.eg